

Ime Asamudo

Portfolio Link | 587-432-3720 | imeahonusiasamudo@gmail.com | <https://www.linkedin.com/in/ime-asamudo/>

Education

MSc in Computer Science

Georgia Institute of Technology

August 2025 – Current

Atlanta, Georgia

BSc in Electrical Engineering

University of Alberta

September 2019 – April 2024

Edmonton, Alberta

Experience

Embedded Linux/Firmware Engineer

TEKTELIC Communications Inc.

June 2024 – September 2025

Calgary, Alberta

- Developed and maintained Kona OS, the embedded Linux platform powering Kona LoRaWAN gateways deployed in 1000+ industrial and smart-city IoT installations.
- Improved CI/CD efforts by migrating TEKTELIC gateway applications to build within the existing Yocto BSP environment, cutting manual build steps by 50% and improving security, reproducibility, and integration.
- Ported U-Boot and Linux kernel to ARM Cortex-A8 based SoCs, ensuring compatibility with custom hardware.
- Developed SPI, I2C, and UART drivers in both kernel and user mode to interface with FPGAs, GPS receivers, and various ASICs, enabling reliable communication between hardware components.
- Developed a library for the ATECC608B secure element and supporting utility in C, supporting ECC operations, ECDSA signature generation, and ECDH key exchange to integrate hardware security into IoT applications.
- Contributed to the Hardware Abstraction Layer (HAL) for a LoRa concentrator by integrating SPI and GPIO support, enabling communication with multiple LoRa devices across 64 different frequency bands.
- Integrated unit testing in the existing code base using the Ceedling Framework improving code quality and allowing for effective Test Driven Development.

Embedded Software Engineering Intern

Intelligent IoT and Ultrasound Lab at University of Alberta

May 2023 – December 2023

Edmonton, Alberta

- Devised a Bluetooth-enabled IoT Proximity Detector capable of monitoring the distance of a probe to a sensor with 1mm sensitivity and 90% accuracy for biomedical applications.
- Engineered C firmware for a wireless application on a ARM Cortex-M0 based micro-controller, leveraging FreeRTOS to optimize ADC sampling and BLE data transmission, resulting in a 60% reduction in latency.
- Developed real-time data logging and visualization software using a multi-threaded C# .NET WinForms framework, incorporating digital signal processing techniques like a moving average filter to achieve a 40% improvement in data processing efficiency, resulting in a highly responsive graphical user interface(GUI).
- Integrated hardware components by creating schematics and the layout for 3 2-layered PCBs using Altium Designer, followed by hands-on soldering and the use of a reflow oven.

Projects

Causality SoC

May 2025 - August 2025

- Implemented an ARM Cortex-M0 SoC on the Xilinx Artix-7 FPGA in Verilog and developed C drivers to make a HAL to interface on-chip peripherals and run custom applications.

BLE ECG Logger

January 2024

- Developed a biophysical instrumentation device that captures the electrical signal of a user's heart and transmits real-time ECG data via BLE to a PC-based user interface for wireless monitoring.

Technical Skills

Programming Languages: C/C++, Bash, Python, C#, Verilog, VHDL, ARM Assembly

Technologies: Architectures(ARM Cortex-M, ARM Cortex-A), Debugger(GDB, SEGGER J-Link, STM ST-LINK/V2), FPGA(Xilinx, Lattice)

Protocols: SPI, I2C, UART, USB, TCP/IP, UDP/IP, LoRaWAN, BLE

Developer Tools: Git, Jira, Confluence, Visual Studio Code, STM32CubeIDE, Keil uVision5, LTspice, Altium Designer, KiCad, Xilinx Vivado, MATLAB/SIMULINK, Pathwave (ADS)

Hardware Tools: Oscilloscope, Logic Analyzer, Function Generator, Vector Network Analyzer, Multimeter